King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 252

Original exam date: 17 February 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The product of all possible values of α such that the points

$$P(0,2,1),\ Q(\alpha-1,0,\alpha),\ R(5,-\alpha,6)$$

are collinear, is equal to

(a) -12

(b) 12

(c) 1

(d) 7

(e) -1

2. The polar equation $r^2 = 4\sec\theta\csc\theta$ can be converted to the rectangular equation

(a) $y = 2x$

(b) $y = \dfrac{4}{x}$

(c) $y = \dfrac{x}{2}$

(d) $y = \dfrac{2}{x}$

(e) $y = 4x$

3. The vector component of $\vec{u} = \langle -9,-2,-4\rangle$ orthogonal to $\vec{v} = \langle 0,4,4\rangle$ is

(a) $\vec{u} - \dfrac{4}{3}\vec{v}$

(b) $-\dfrac{3}{4}\vec{v}$

(c) $\dfrac{3}{4}\vec{v}$

(d) $\vec{u} + \dfrac{3}{4}\vec{v}$

(e) $\vec{u} - \dfrac{3}{4}\vec{v}$

4. The area of the region enclosed by the cardioid $r = 2(1 + \cos\theta)$ is equal to

(a) 2π

(b) 12π

(c) 3π

(d) 4π

(e) 6π

5. The graph of $(x^2 + y^2)^3 = 4x^2y^2$ is

(a) a limacon with an inner loop

(b) a lemniscate

(c) a cardioid

(d) a circle

(e) a four-petal rose

6. Let α be the angle between the two nonzero vectors $\vec{u}$ and $\vec{v}$. If

$$\vec{u} \cdot \vec{v} = -1 \quad \text{and} \quad \|\vec{u} \times \vec{v}\| = \frac{1}{\sqrt{3}},$$

then $\sec(\alpha) =$

(a) $\sqrt{3}$

(b) -2

(c) $\dfrac{2}{\sqrt{3}}$

(d) $-\dfrac{2}{\sqrt{3}}$

(e) $-\sqrt{3}$

7. The triangle with the vertices

$$P(1, 2, 0),\ Q(0, 0, 0),\ R(-2, 1, 0)$$

(a) is an obtuse triangle

(b) has an area of 5

(c) has an area of 10

(d) is a right triangle

(e) is an acute triangle

8. If the standard equation of the sphere with center $(-4, 5, 6)$ and tangent to the yz-plane is $(x-a)^2+(y-b)^2+(z-c)^2=r^2$, then $a+b+c+r^2=$

(a) 23

(b) 31

(c) 32

(d) 43

(e) 11

9. The graph of the parametric curve

$$x=\frac{4}{\cos t}, \quad y=\frac{3\sin t}{\cos t}$$

is

(a) a hyperbola

(b) a parabola

(c) a line

(d) a circle

(e) an ellipse

10. If the area of the surface generated by revolving the parametric curve

$$x=\frac{t^3}{3}, \quad y=t+1, \quad 1\le t\le 2,$$

about the y-axis, is equal to $\frac{\pi}{9}\left(a^{3/2}-b^{3/2}\right)$, then $a+b=$

(a) 19

(b) 15

(c) 17

(d) 18

(e) 16

11. The parametric curve

$$x=t-6, \quad y=t^2$$

(a) is concave upward for all values of $t\in(-\infty,\infty)$.

(b) has a slope 2 when $t=-1$.

(c) is concave upward only for $t>0$ and concave downward for $t<0$.

(d) has a horizontal tangent at $t=1$.

(e) is concave downward for all values of $t\in(0,\infty)$.

12. If the angle between the two unit vectors $\vec{A}$ and $\vec{B}$ is $120°$, then $\|\vec{B} - \vec{A}\| =$

(a) 1

(b) $\sqrt{3}$

(c) 3

(d) 2

(e) $\sqrt{2}$

13. The value of c for which the three vectors

$$\vec{u} = \langle 1, c, -7 \rangle, \quad \vec{v} = \langle 2, -1, c \rangle, \quad \vec{w} = \langle 0, -9, 18 \rangle$$

are coplanar (lie in the same plane) is

(a) $-\dfrac{12}{5}$

(b) $\dfrac{27}{7}$

(c) $\dfrac{1}{4}$

(d) 4

(e) $-\dfrac{2}{3}$

14. The slope of the tangent line to the polar curve

$$r = 2(1 - \cos\theta)$$

at the point corresponding to $\theta = \dfrac{4\pi}{3}$ is equal to

(a) 0

(b) $\dfrac{1}{2\sqrt{3}}$

(c) $\dfrac{1}{\sqrt{3}}$

(d) $\dfrac{\sqrt{3}}{2}$

(e) $\sqrt{2}$

15. The area of the region inside the circle $r = 1$ and outside of the cardioid $r = 1 - \cos\theta$ is equal to

(a) $1 - \dfrac{\pi}{4}$

(b) $4 - \dfrac{\pi}{2}$

(c) $2 - \dfrac{\pi}{4}$

(d) $1 - \dfrac{\pi}{8}$

(e) $2 + \dfrac{\pi}{4}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\overrightarrow{PQ} \parallel \overrightarrow{PR}$ gives $\alpha^2 + \alpha - 12 = 0$, so $\alpha = -4, 3$; product -12.
2	**(b)**	$r^2 \sin\theta \cos\theta = 4$ means $xy = 4$, so $y = 4/x$.
3	**(d)**	proj $= \frac{-24}{32}\vec{v} = -\frac{3}{4}\vec{v}$, so the orthogonal part is $\vec{u} + \frac{3}{4}\vec{v}$.
4	**(e)**	$A = \frac{1}{2}\int_0^{2\pi} 4(1+\cos\theta)^2 d\theta = 6\pi$.
5	**(e)**	$r^2 = \sin^2 2\theta$, so $r = \sin 2\theta$: a four-petal rose.
6	**(d)**	$\tan\alpha = -\frac{1}{\sqrt{3}}$, α obtuse ($150°$), so $\sec\alpha = -\frac{2}{\sqrt{3}}$.
7	**(d)**	$\overrightarrow{QP} \cdot \overrightarrow{QR} = 0$: right angle at Q; area is $\frac{5}{2}$.
8	**(a)**	$(a, b, c) = (-4, 5, 6)$, $r = \lvert-4\rvert = 4$, so the sum is $-4 + 5 + 6 + 16 = 23$.
9	**(a)**	$\sec t = x/4$, $\tan t = y/3$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$, a hyperbola.
10	**(a)**	$S = 2\pi \int_1^2 \frac{t^3}{3}\sqrt{t^4+1}\, dt = \frac{\pi}{9}(17^{3/2} - 2^{3/2})$, so $a + b = 19$.
11	**(a)**	$dy/dx = 2t$, $d^2y/dx^2 = 2 > 0$ for all t: concave upward everywhere.
12	**(b)**	$\lVert\vec{B} - \vec{A}\rVert^2 = 1 + 1 - 2(-\frac{1}{2}) = 3$, so the length is $\sqrt{3}$.
13	**(d)**	det $= -27c + 108 = 0$, so $c = 4$.
14	**(a)**	$r = 3$, $r' = -\sqrt{3}$: numerator $\frac{3}{2} - \frac{3}{2} = 0$, denominator $2\sqrt{3}$, so slope 0.
15	**(c)**	Meet at $\theta = \pm\frac{\pi}{2}$: $A = \int_0^{\pi/2}(2\cos\theta - \cos^2\theta)d\theta = 2 - \frac{\pi}{4}$.